

SQL Reference & Syntax Guide

Comprehensive Quick-Reference Guide for Operators, Clauses, and Queries

1. SQL Arithmetic Operators

Arithmetic operators perform mathematical operations on numeric data values within SQL queries.

OPERATOR	DESCRIPTION	EXAMPLE SYNTAX
+	Addition	<code>SELECT price + tax FROM products;</code>
-	Subtraction	<code>SELECT price - discount FROM products;</code>
*	Multiplication	<code>SELECT quantity * unit_price FROM order_items;</code>
/	Division	<code>SELECT total_amount / item_count FROM orders;</code>
%	Modulo (Remainder)	<code>SELECT product_id % 2 FROM products;</code>

2. SQL Comparison Operators

Comparison operators test conditions in the `WHERE` clause to filter records.

OPERATOR	MEANING	EXAMPLE QUERY
=	Equal to	<code>SELECT * FROM employees WHERE dept = 'IT';</code>
<> or !=	Not equal to	<code>SELECT * FROM users WHERE status != 'Active';</code>
> / <	Greater than / Less than	<code>SELECT * FROM products WHERE price > 100;</code>
>= / <=	Greater/Less or equal	<code>SELECT * FROM orders WHERE qty <= 5;</code>

3. Special SQL Operators

BETWEEN Operator

Selects values within an inclusive range (numbers, text, or dates).

```
SELECT * FROM products
WHERE price BETWEEN 10 AND 50;
```

LIKE Operator

Searches for a specified pattern in a column using wildcards (`%` for multiple chars, `_` for single char).

```
-- Matches any customer name starting with 'A'
SELECT * FROM customers
WHERE customer_name LIKE 'A%';
```

IN Operator

Allows specifying multiple values in a `WHERE` clause as a shorthand for multiple `OR` conditions.

```
SELECT * FROM employees
WHERE department IN ('HR', 'IT', 'Finance');
```

4. Filtering & Unique Results

WHERE Clause: Filters rows based on conditional expressions.

```
SELECT employee_id, first_name, salary
FROM employees
WHERE salary >= 50000 AND department_id = 10;
```

SELECT DISTINCT Clause: Returns only unique, non-duplicate values.

```
SELECT DISTINCT country FROM customers;
```

5. ORDER BY Clause (Sorting Results)

Sorts the result set in ascending (`ASC` , default) or descending (`DESC`) order.

Single Column Sort:

```
SELECT * FROM products
ORDER BY price DESC;
```

Multiple Column Sort:

```
SELECT * FROM employees
ORDER BY department_id ASC, salary DESC;
```